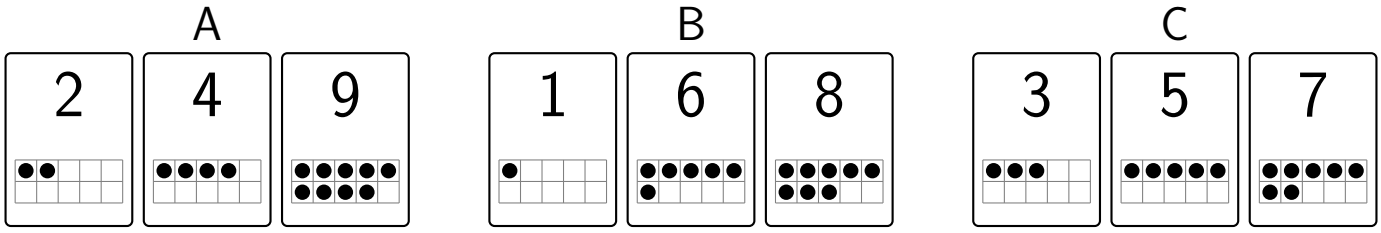
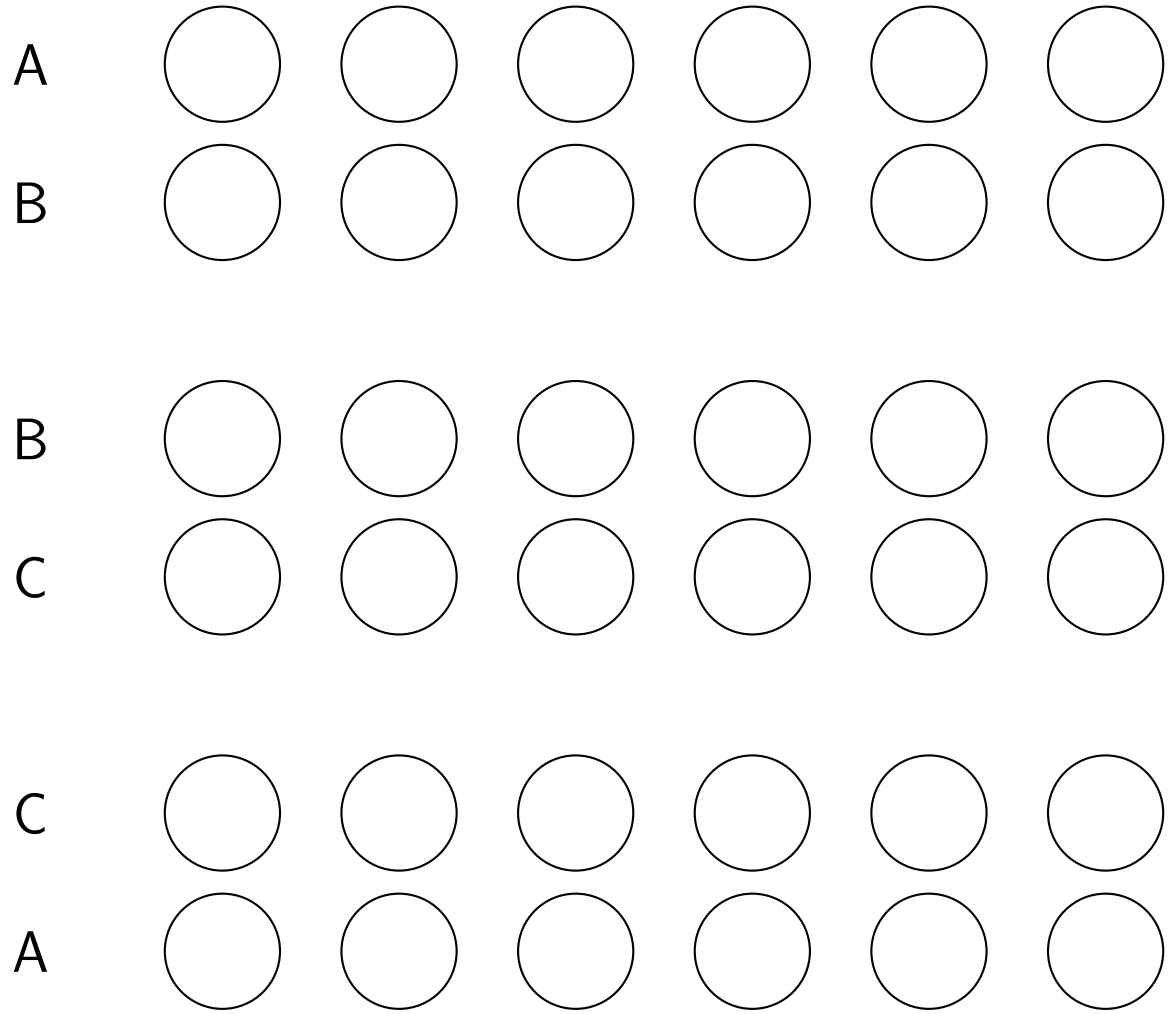
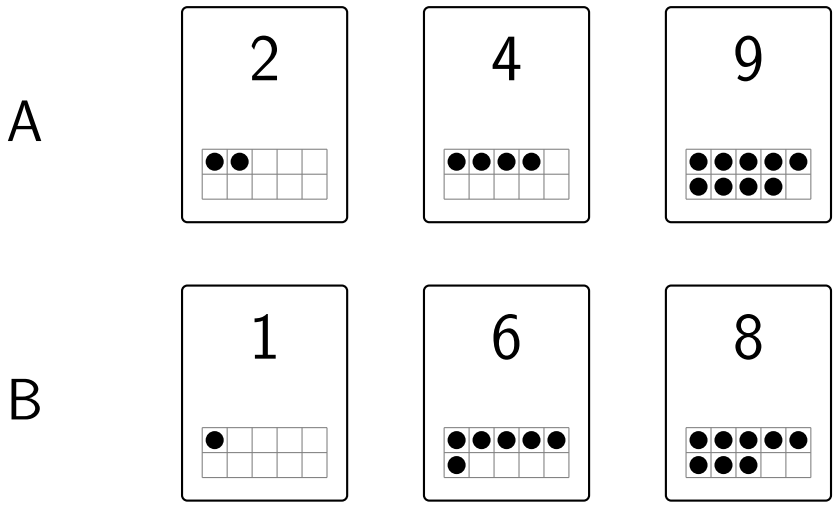
Decks A, B and C are bags of three number cards, shown below. Mix each bag on its own. In a game, each player takes one deck. Each round, draw one card from each player's bag without looking; the bigger number wins the round. Put both cards back before the next round.

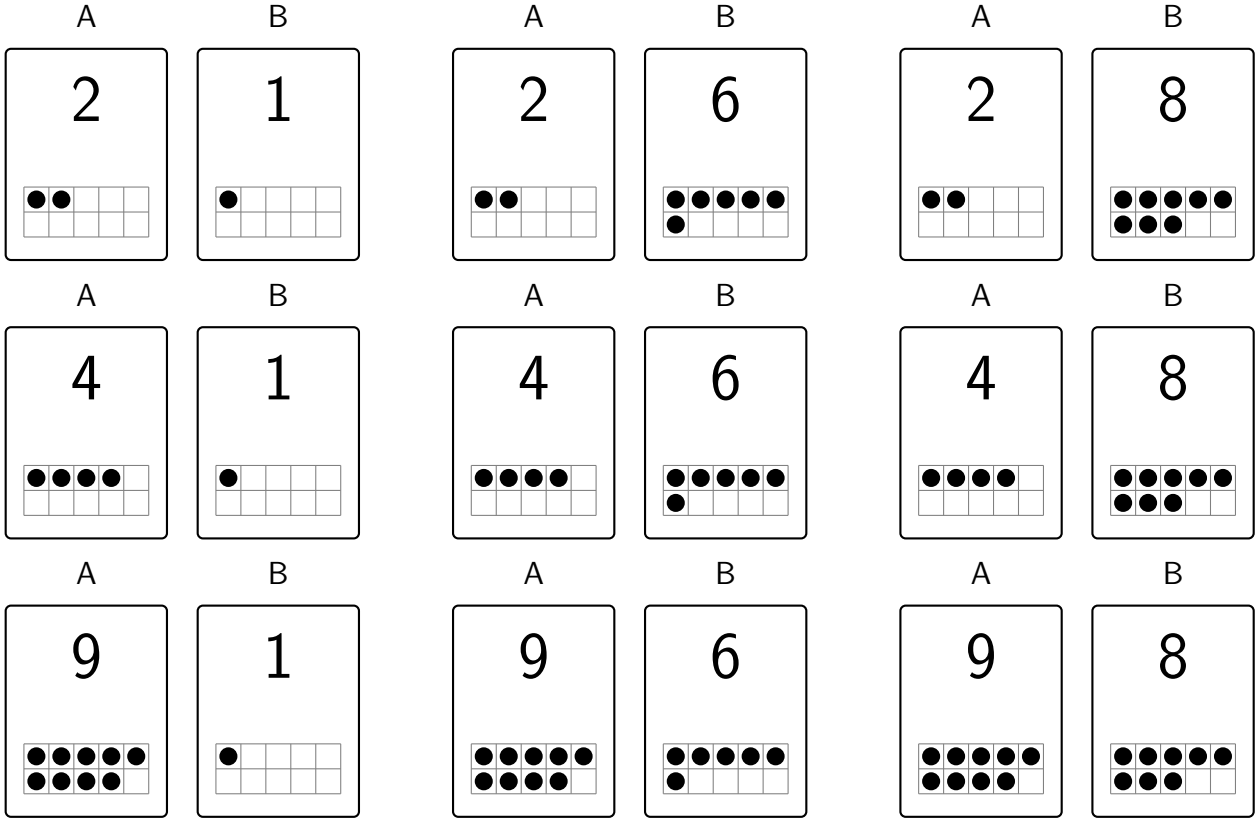


Problem 1: With a partner, play three games: A against B, B against C, and C against A. Play six rounds in each game, and put a counter in the winning deck's row each round. Is one deck better than both of the others?

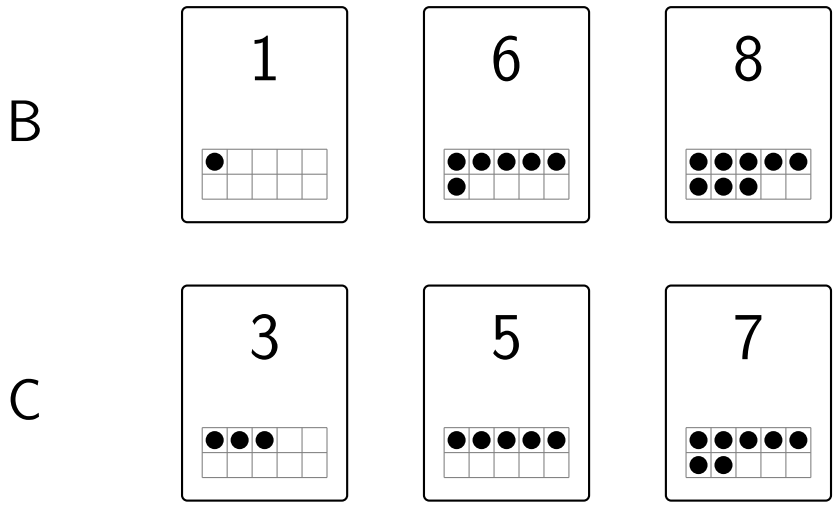


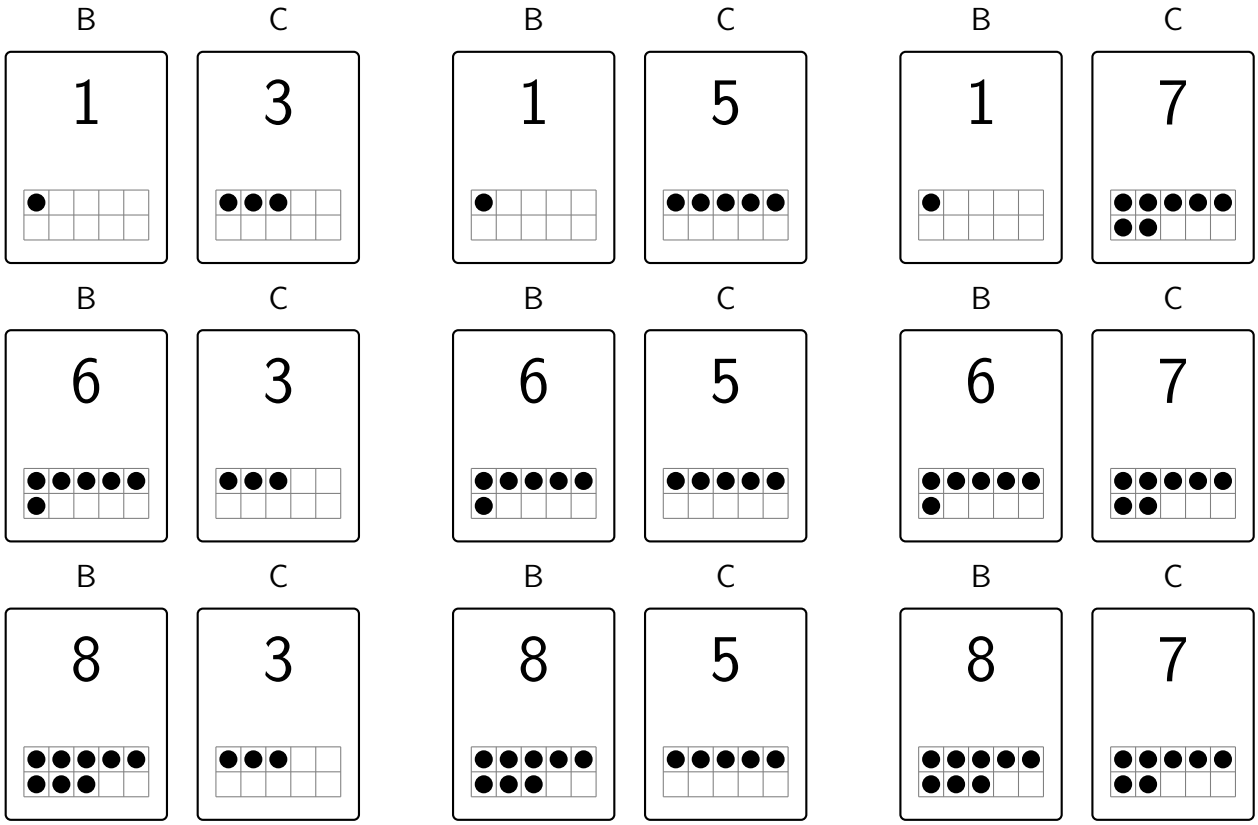
Problem 2: Circle the winner in each pair. Which deck wins more pairs?



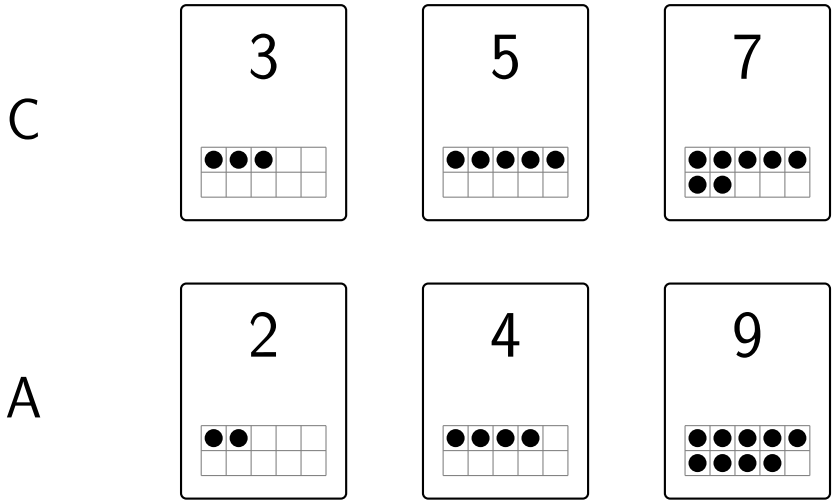


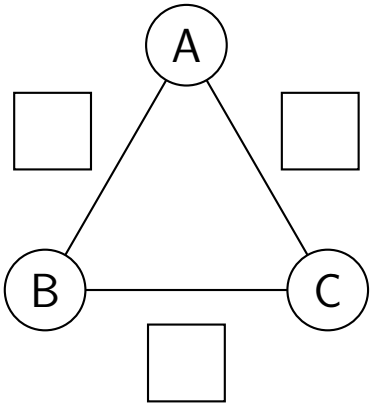
Problem 3: Circle the winner in each pair. Which deck wins more pairs?

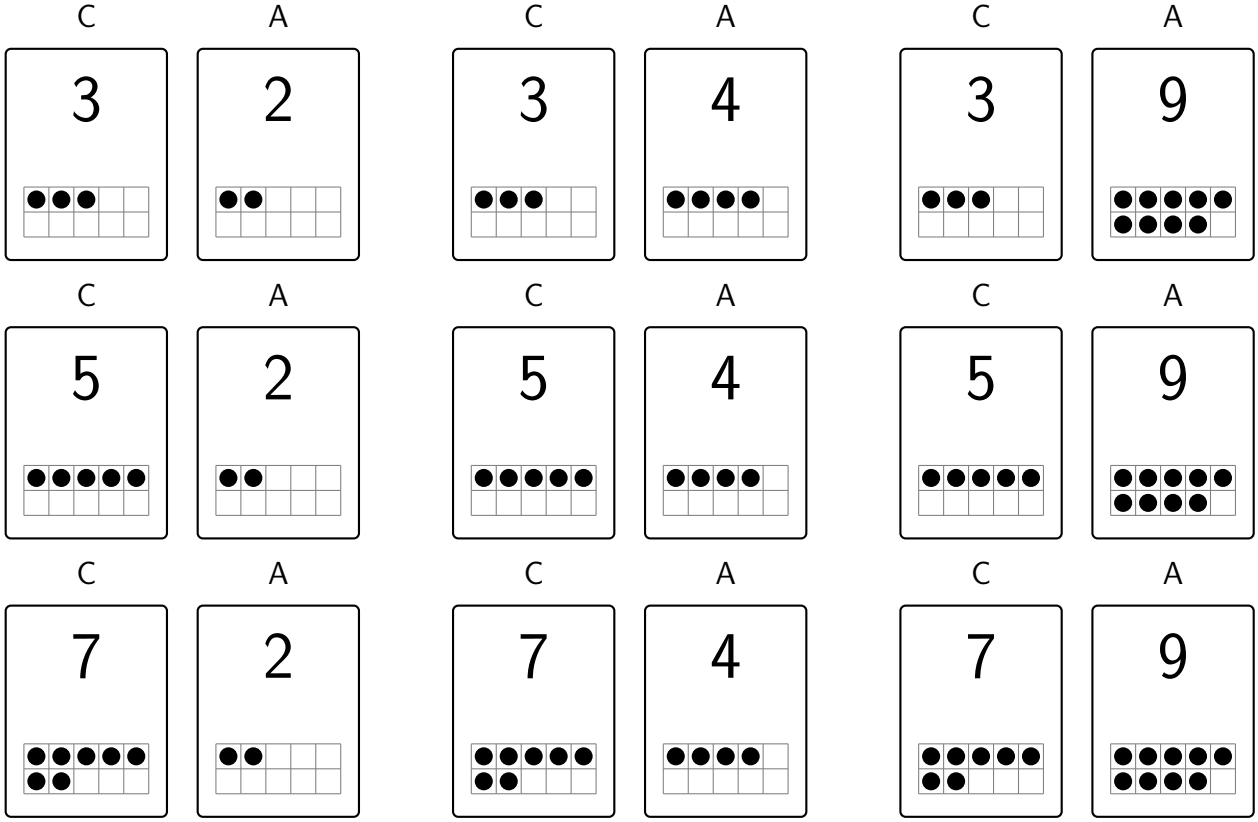




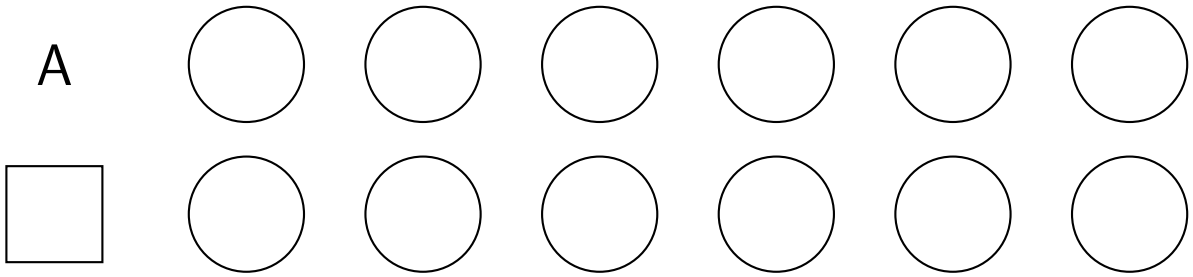
Problem 4: Circle the winner in each pair. In each box, write the deck that wins more pairs. Does any deck win more pairs than both other decks?

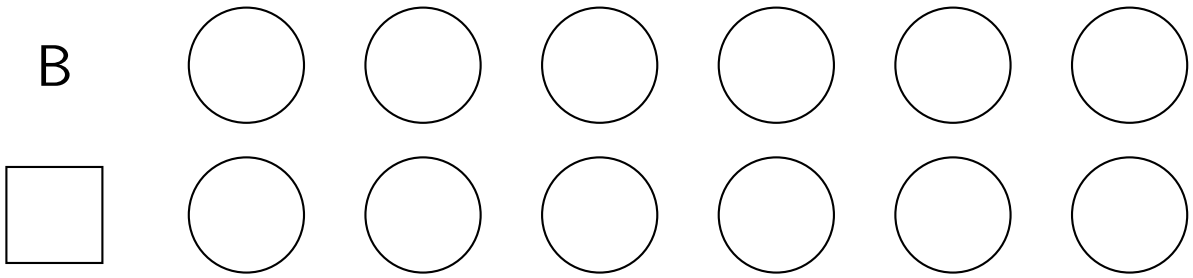


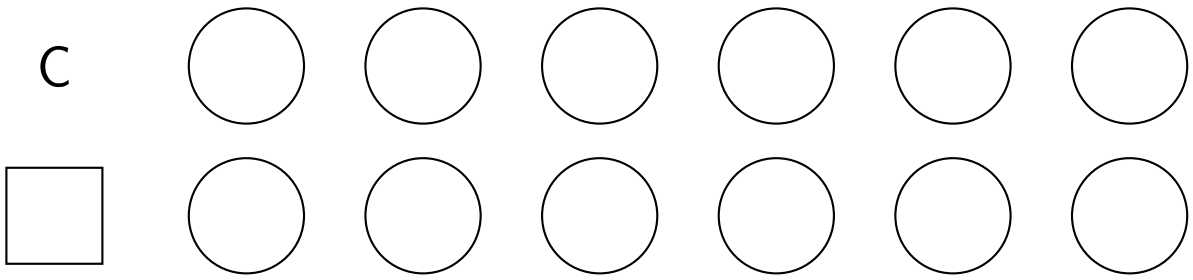




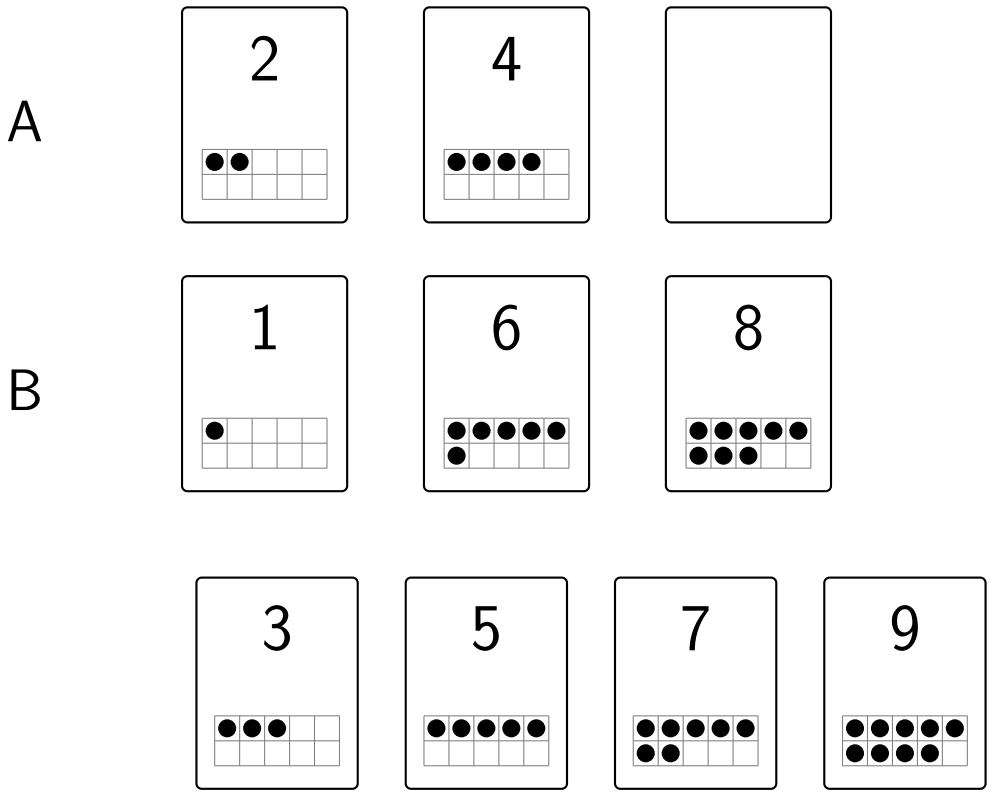
Problem 5: Your partner takes A. Choose a deck that wins more card pairs against it, write its letter in the box, and play six rounds. Do the same for B and C. Must your chosen deck win more rounds?



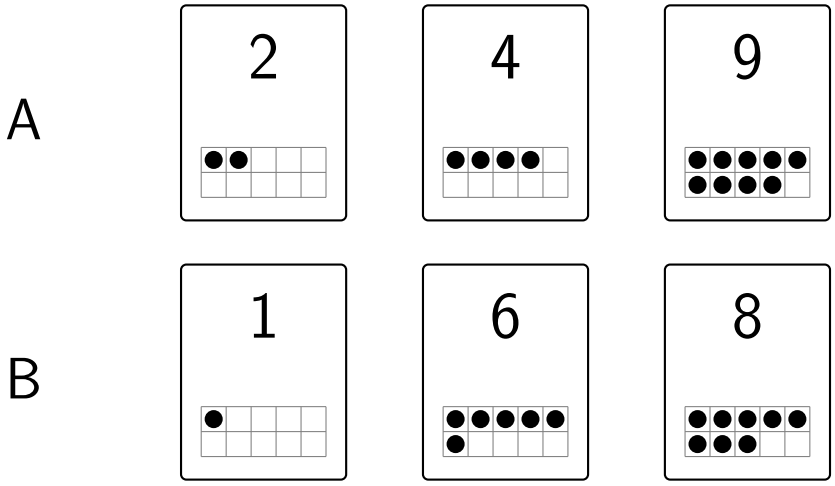


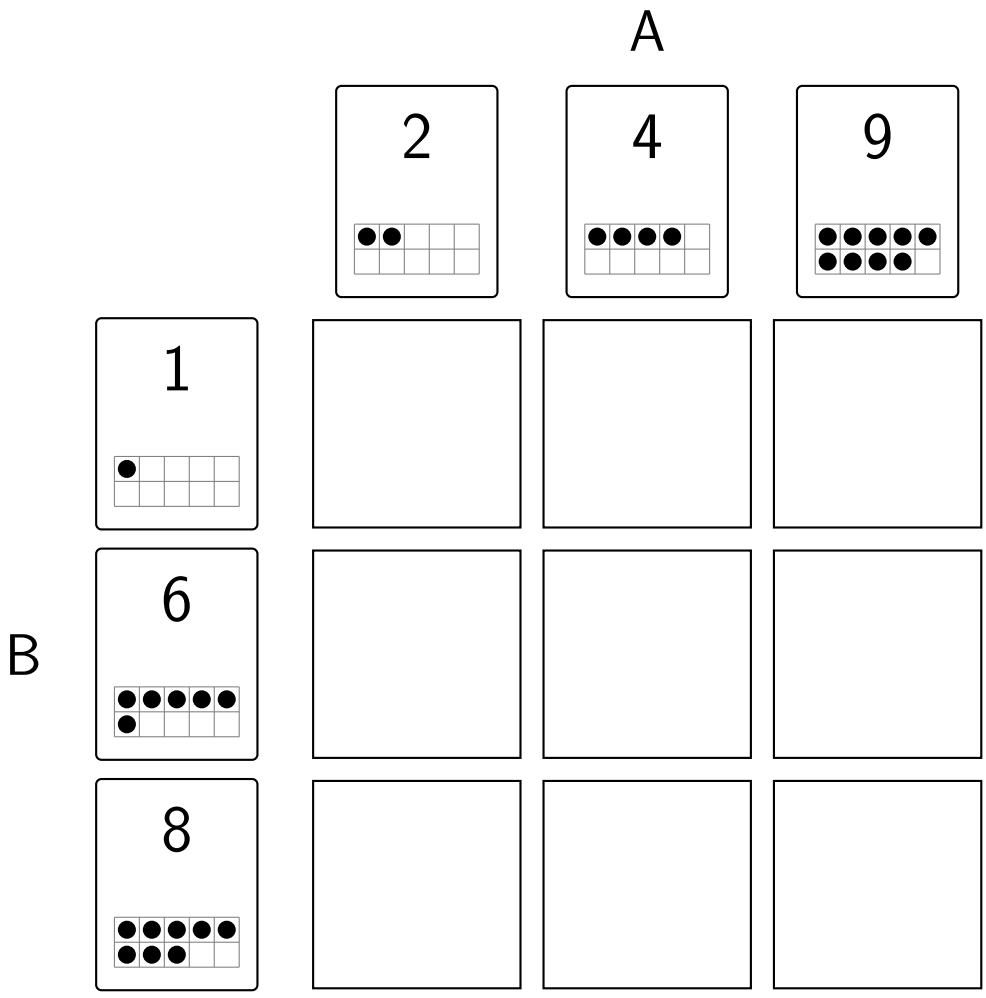


Problem 6: Try 3, 5, 7, and 9 in A's empty place. Which choices make A win more card pairs than B?

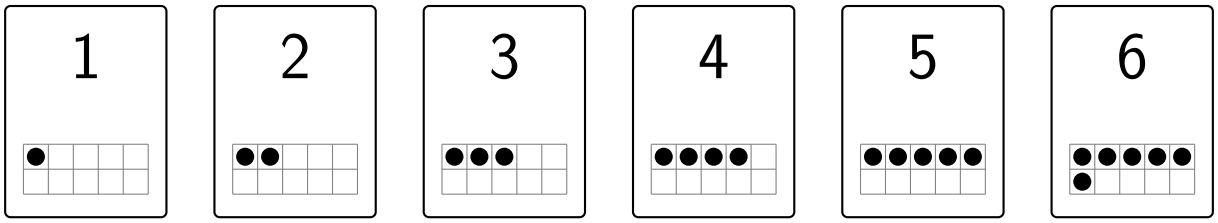


Problem 7: Swap one A card with one B card, starting from these decks each time. Mark the box of every swap that makes B win more card pairs than A.





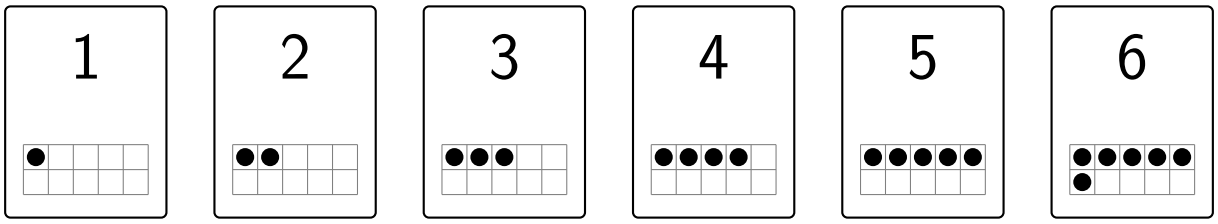
Problem 8: Put these six cards into two decks of three so both decks win the same number of pairs. Find a way, or show why it cannot be done.



A

B

Problem 9: Can these six cards make three two-card decks where A wins more pairs than B, B more than C, and C more than A? Show a way, or show why it cannot be done.



A

B

C